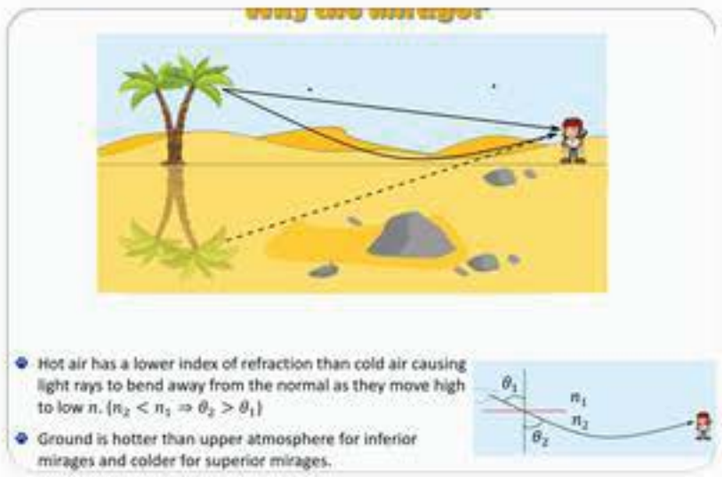
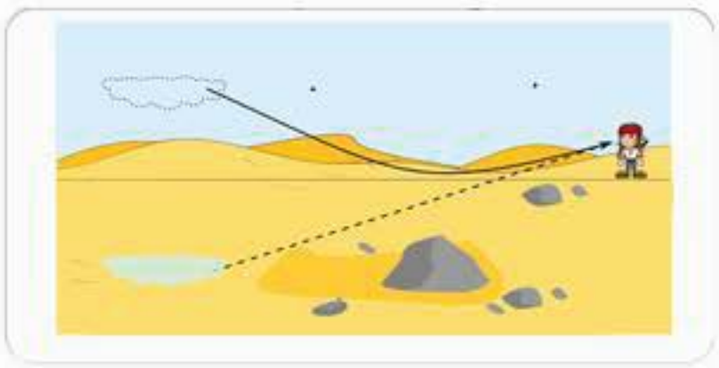
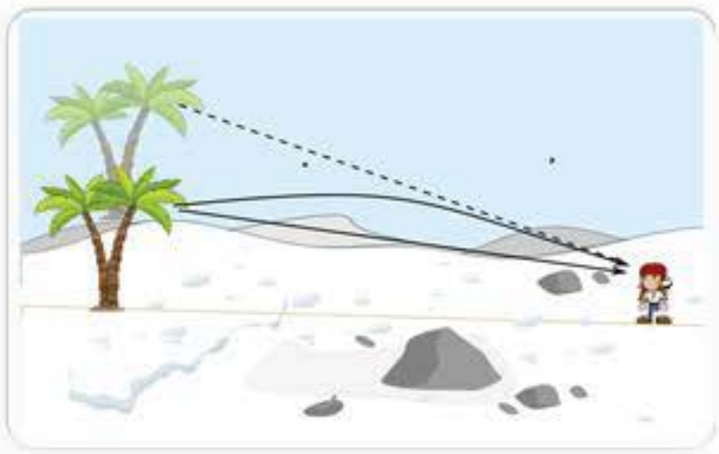


The photos are from my slide deck. I made them a few years back. I believe I still have the SVGs if you need them. Or we might still use our beloved clown. The first two images focus on the regular mirage, the last on the superior mirage. I hope that we make a SIM that explores this, may be allowing for the gradient in n to be changed.

Thought for 45s

A mirage sim is a great follow-on to the refraction work. Your slides already show the key picture: curved rays, and the straight dashed line back to where the eye thinks the object is. The clown will work well as the object; if you'd like your palm trees and desert art from the SVGs later, they can replace it. Here's the design.

The physics: air whose temperature, and so whose index n, varies with height.

- **The ground's temperature** differs from the air above by ΔT, fading over a layer of thickness h.
- **The index follows density:** $n - 1 \propto 1/T$, so hot air has the lower n, as your slide says.
- **Each ray is traced exactly** through the layers, using Snell's law in its continuous form: $n \cos \alpha$ stays constant along a ray, where α is the angle from the horizontal.

Three presets, then free sliders for ΔT, h, the distance and the eye height:

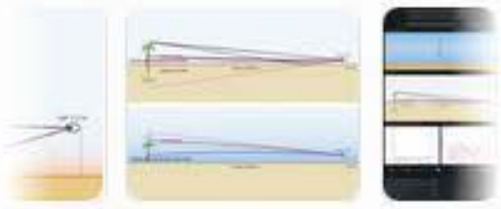
- **Hot ground** (desert, road): the inferior mirage, an inverted image below the real one, and the "water" on the road, which is the sky's image.
- **Cold ground** (cold sea, snow): the superior mirage, the object raised (looming), as on your third slide.
- **Uniform air:** no mirage, for comparison.

The views:

- **A side view with the heights exaggerated,** as in your slides: the real, curved paths from the clown to the eye, and the dashed straight lines to where the eye sees each image. The air is shaded by temperature.
- **What the observer sees,** rendered by tracing a ray for every pixel: the clown, plus its mirage, inverted below or raised above.
- **The temperature and the index against height,** with the eye's height marked.

Let me check that the scratch files from the refraction sim (its styles and the compact clown) still exist:

Ran 10 commands, read 5 files, created a file



The mirage sim is ready as `go-mirage-inferior-superior`, with GitHub and MOM versions and a SIM description. It's in `sims-mobile.zip`, and the inventory lists 134 sims.

The physics: the ground's temperature differs from the air above by ΔT, fading with height over a layer of thickness h. The index follows the air's density ($n - 1 \propto 1/T$), so hot air has the lower index, as on your slide. Every ray is traced exactly through the layers, by Snell's law in its continuous form, $n \cos \alpha$ = constant along each ray; I checked that this holds to rounding error.

Four presets, then free sliders for ΔT (−20 to +40 K), h, the distance (up to 3 km) and the eye's height, and the clown or a palm tree as the object:

- **Hot road:** the clown at 900 m, seen from a driver's eye height. The inferior mirage is an inverted clown below the real one, floating in a band of sky that looks like water on the road.
- **Desert oasis:** a palm tree at 1.5 km with its inverted twin below, your first slide.
- **Cold sea:** the palm raised above where it really is (looming), the superior mirage of your third slide.
- **Uniform air:** straight lines, no mirage.

The views:

- **What the observer sees,** rendered by tracing a ray for each row of pixels. It's a telephoto view, about 30–60 arc minutes tall, since mirages are only a fraction of a degree in size.
- **A side view with the heights exaggerated,** like your slides:
  - *The air,* shaded by temperature.
  - *The real light paths,* curved, from the object's top and lower part to the eye.
  - *The dashed straight lines* along which the eye sees each image, with the image drawn faintly where the eye places it: the inverted one below the ground, the raised one above.
- **Temperature and index against height,** with the eye's height marked.
- **The height each viewing angle reaches at the object.** For an inferior mirage the curve is U-shaped, so two viewing angles reach the same point of the object, which is why it's seen twice.

The readout gives the index gradient at the ground, the radius of a grazing ray (about 15–30 km over a hot road), the bending over the distance, and the angle of each image.

What the sim teaches by itself: with the original settings I tried (an eye at 1.7 m and a clown at 600 m), there was **no** mirage, and the sim said so: "At this distance the bending is too small for a mirage: move the object farther, or the eye lower." That's real physics, and why road mirages appear in the distance and to a low eye.

Two notes:

- **I used the clown and a palm tree I drew.** If you send your SVGs, I can use your palm trees and your desert scenery for the real objects.
- **The ground is treated as flat.** The Earth's curvature matters for the most dramatic superior mirages (ships seen from beyond the horizon), and could be added later.

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While working on this, I was verifying the snell's lab SIMs. They work great, but the uncertainty in the slope is missing. Can you please add it.

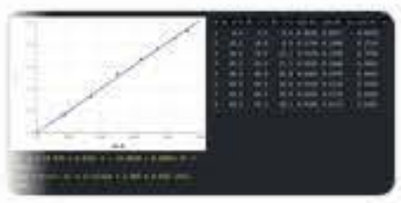
Thought for 5s

The slope's uncertainty belongs in any honest fit, and it should carry through to the index students compute from it. I'll use the standard least-squares formulas:

- **For the free fit,** the residual variance is $s^2 = \Sigma(\text{residuals})^2/(n-2)$, and then $\sigma_m = s/\sqrt{Sxx}$ and $\sigma_b = s\cdot\sqrt{(1/n + \bar{x}^2/Sxx)}$, where $Sxx = \Sigma(x-\bar{x})^2$. This needs at least 3 runs.
- **Through the origin,** $s^2 = \Sigma(\text{residuals})^2/(n-1)$, and $\sigma_m = s/\sqrt{\Sigma x^2}$. This needs at least 2 runs.
- **The index from the slope gets its own uncertainty:** $\sigma_n = n_1\sigma_m/m^2$ when $n_2 = n_1/\text{slope}$, and $\sigma_n = n_1\sigma_m$ when $n_2 = \text{slope} \times n_1$.

The fit line will read, for example, "$y = (0.6682 \pm 0.0035)\ x - (0.012 \pm 0.004)$", and the interpretation "$n_2 = 1.497 \pm 0.008$". I'll patch the latest versions in the library, which have your edits and the saved-data feature. Let me find the code to change:

Ran 5 commands, read a file, shared files



Both Snell's law labs now report the uncertainty of their fits, in GitHub and MOM versions, in `sims-mobile.zip`. Your edits and the saved-data feature are unchanged.

What the fit now shows:

- **The slope and the intercept, each with its standard error:** for example "fit: $y = (0.675 \pm 0.015)\ x - (0.0018 \pm 0.0091)$", each uncertainty rounded to two significant figures and its value to match.
  - *Free fit:* the standard least-squares errors, with $n - 2$ degrees of freedom. It needs at least 3 runs, and says so with fewer.
  - *Through the origin:* the slope's error with $n - 1$ degrees of freedom, from 2 runs.

- **The index from the slope, with its propagated uncertainty:** for example "$n_2 = 1.481 \pm 0.032$ (set: 1.50)". The uncertainty is $n_1\sigma_m/m^2$ where $n_2 = n_1/\text{slope}$, and $n_1\sigma_m$ where $n_2 = \text{slope} \times n_1$.
- **For the two graphs whose slope should be 1** (Snell's law as n sin θ, and the law of reflection), whether the measured slope is consistent with 1: "your slope (1.013 ± 0.022): consistent with 1 (within 1 standard error)", or by how many standard errors it differs.

Checked: the slopes, intercepts and their uncertainties match an independent Python calculation, for both fits in both labs. Both labs work on a 360 px phone, and their data still survive a reload.

A teaching opportunity the uncertainties reveal: the protractor's systematic misalignment, part of the experimental error, can push a fit *through the origin* away from the truth by several standard errors. In one inverted-lab run, it gave $n_2 = 1.559 \pm 0.017$ against the true 1.50, while the free fit gave 1.512 ± 0.027, consistent with it. The free fit absorbs the systematic offset in its intercept (0.021 ± 0.010, itself not consistent with zero). It's a good concrete case for students on when to force a fit through the origin, and on how a systematic error can hide behind a small statistical one.

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Go snells law inverted half disk refractions lab mom
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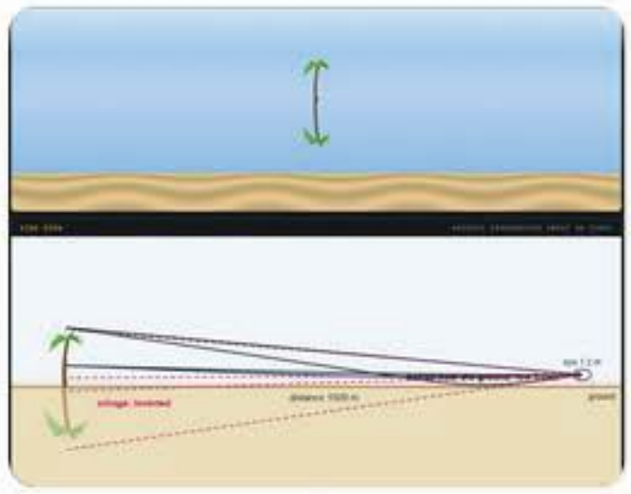
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Let us focus now on the mirage. Very good start. (1) We have an misunderstanding, I meant for the clown to replace the boy as the viewer instead of the eye you are using. (2) Having the clown as the object to be viewed is a little tricky and I am not sure if it ever happens in real life. Mirages of trucks or cars on hot roads are common. As the attached image also shows, the mirage also shows puddles of water in the road (it happens in deserts too) which is something I hope we will address (3) I am not sure I understand what is shown in the "WHAT THE OBSERVER SEES" panel. The observer should not see the palm tree along with its mirage floating. (4) This graph "TEMPERATURE AND INDEX AGAINST HEIGHT" is not clear missing labels for the x axis. (Is there too much focus on the height where the change has ceased to happen?)

Thought for 47s

Thank you for the careful review. All four points are fair. Here's what I'll change.

1. The clown becomes the observer, standing at the right of the side view with his eyes at the eye height (the "eye height" slider then sets how tall he stands). That's your slides' boy, and our beloved clown.

2. The objects become real ones:

- **A hot road** with a truck on it, as in your photo, and telephone poles along the road at intervals.
- **A desert** with palm trees.
- **A cold sea** with a ship, for the superior mirage.

The "puddle" on the road is the sky's own mirage, light from the sky bent up off the hot layer. With a real road drawn in perspective (asphalt, lane markings, the poles receding), it will look like water, and I'll label that band in the view.

3. The observer's view. You're right that it was confusing. The problem was that the object was floating in empty sky: the view showed nothing else to judge it by. Your photo shows what's missing: the road receding to the horizon, the poles, and the puddle lying on the road. I'll render that full scene in perspective, with a zoom setting, since real mirage photos are telephoto shots too. The truck will then sit on the shimmering road with its inverted image in the puddle below it, as in your photo.

4. The temperature-and-index graph:

- **Labelled axes:** temperature (°C) along the bottom and n − 1 (in millionths) along the top.
- **Its height range** focused on the layer where the change actually happens, rather than mostly empty air above it, with the eye's height still marked.

Let me check whether my working copy of the sim survived, since the scratch space gets cleared:

Ran a command

Continue

Ran 6 commands, read 3 files, created a file · 1 note



The mirage sim is reworked on all four points, in `go-mirage-inferior-superior.html`, its MOM version, and `sims-mobile.zip`.

1. The clown is the observer. He stands on the right of the side view, with his eyes at the eye height. The slider is now "height of the clown's eyes", and the top panel is "What the clown sees".

2. Real objects and the "water":

- **A hot road** with a truck (in the right lane, 1.1 km away), telephone poles along both sides and lane markings. Seen through the hot air, the far road turns into a band of "water", light from the sky bent up by the hot layer, labelled in the view. The truck stands in it with its upside-down image below it, and the poles continue into it as reflections, much like your photo.
- **The desert:** palm trees standing in an "oasis".
- **A cold sea:** a ship, raised by the superior mirage.
- **The same road in air of one temperature:** a dry road, and everything seen once, for comparison.

The object menu offers a truck, a car, a palm tree or a ship in any situation.

3. The observer's view is now a whole scene in perspective, the ground receding to the horizon, so it's clear what is real and what is mirage. It's a telephoto view, as mirage photos are, since mirages are only fractions of a degree. A new **zoom** slider (0.4° to 8°) lets students widen it to see more of the scene, or narrow it to study the mirage.

4. The graph, now "the air near the ground":

- **Its height range covers only where the temperature actually changes:** about 4 times the layer's thickness, or a little above the clown's eyes.
- **Its axes are labelled:** temperature (°C) along the bottom and (n − 1) in millionths along the top, with the curves labelled where they're apart.

Unchanged physics: the rays are still traced exactly ($n \cos \alpha = \text{constant}$), and the side view and the angle graph work as before.

One cost to know about: each redraw now takes about 0.15–0.3 s, because every row's ray is traced out to 12 km to find the poles and the horizon. That's fine for a slider; if it feels sluggish on older devices, I can render a draft while dragging and the full view when it stops.

If you send your SVGs, I can use your palm trees and your boy, and your desert art for the scenery.

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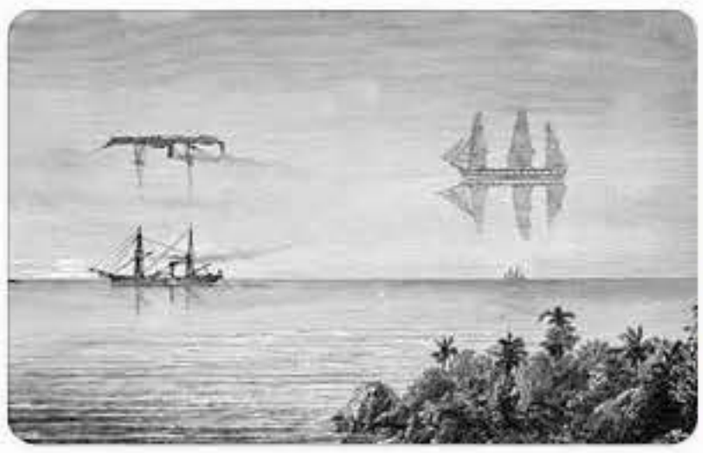
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SVG

tm-mirage01.svg
SVG

tm-mirage02.svg
SVG

Mirage-diagram.svg
SVG

Mirage-Espejismo.svg
SVG

mirage-Superior_and_inferior_mirage.svg
SVG

The files preceded by TM were the ones I worked on back then. The jpg is public domain photo of Primm, Nevada on April 4, 2007. The black and white image has this description: A 19th century book illustration, showing grossly misleading fictional versions of superior mirages. Actual mirages can never be that far above the horizon, and a superior mirage can never increase the length of an object as shown on the right. The last photo, with cars, shows the mirage on Asphalt (and not on the surrounding cooler ground). The other SVGs are SVGs I saved while working on mine. I am hoping that the simulation will help explain what is happening better than stills. For example, the diagrams showing the rays pointing to the directly viewed image as well as the mirage are useful. Something showing how the gradient in n results in the bending will be good. It will be also good to have one example where we can view the fake lake or puddle without anything else in the view (like the nevada photo). The boat and the truck, need a little toning down. For the truck, the water should be only on the asphalt.

Thought for 28s

These are excellent references: your own diagrams, the Nevada “lake”, the 19th-century illustration (a good warning about what *not* to draw), and the cars on asphalt. Here’s the plan:

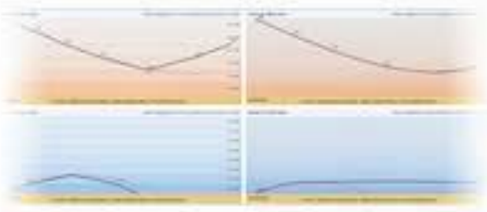
1. **“Why rays bend”**, a new panel that builds the bending from Snell’s law. The air near the ground is cut into layers, each a little hotter than the one above, with lower n. A ray crosses each boundary by Snell’s law, bending a little each time, and at the bottom it’s totally reflected, then climbs back up. A slider sets the number of layers, so students see the steps smooth into the continuous curve, and n cos α stays the same at every boundary.
2. **A “mirage lake” with nothing else in it,** like the Nevada photo: flat desert, distant mountains, and the shimmering “lake” at their feet reflecting the sky and the mountains’ base.
3. **On the road, the water only on the asphalt.** In reality only the asphalt is hot; the shoulders are cooler. So each ray will be traced over the road’s hot air or the shoulders’ cooler air, depending on where along its path it comes closest to the ground. The “water” then appears only on the road, as in your cars photo.

4. **Toning down the truck and the ship:** smaller, softer colours, and more haze. Nothing as dramatic as the 19th-century drawing.
5. **Your art:** let me look at your SVGs, to use your palm trees (and anything else that fits) in place of the ones I drew.

Ran 3 commands, read a file · 1 note

Continue

Those are broken-image icons: a page created in memory isn't allowed to load local files. I'll write the wrapper page into a file next to the drawings instead:



Ran 15 commands, read 9 files, shared files

The mirage sim is updated with everything you asked for, in `go-mirage-inferior-superior.html`, its MOM version, and `sims-mobile.zip`.

How the gradient in n bends the rays: a new panel, "Why rays bend: layers of air".

- **The air is cut into layers,** each a little hotter, and of lower index, toward hot ground. The ray crosses each boundary by Snell's law, its angle from the normal growing at each one (66° → 67° → 70° → 75° …), until it can't enter the next layer and is totally reflected back up.
- **A slider sets 2 to 80 layers,** so students watch the steps merge into the smooth curve, which can be shown dashed. n sin θ is displayed, the same at every boundary.
- **Over cold ground, the panel shows the reverse:** a ray leaving the ground upward bends back down, the superior mirage.
- **The indices are exaggerated so the bending shows,** and the panel says so: in real air they differ by about 2 parts in 100 000.

The "lake" by itself, like the Nevada photo: a new "dry lake bed" situation, with a flat playa, low hazy mountains 30 km away and nothing else. With hot ground, a shimmering bluish lake appears at the foot of the mountains, reflecting their lower slopes and the sky. With uniform air, the desert simply meets the mountains. The object menu also offers "nothing", so any scene can be viewed empty.

On the road, the water is only on the asphalt. Only the asphalt is hot; the shoulders keep a fifth of its excess temperature. Each pixel uses the air the light actually passed through, according to where along its path it came closest to the ground. So the "water" lies on the road, and the dry shoulders and the poles beside them aren't mirrored, as in your cars photo. A checkbox, "shoulders cooler than the asphalt", lets students make them as hot, and watch the water spread across them.

Toned down: the truck is now a muted grey-beige and slightly smaller, and the ship is smaller and greyer. Distant objects are hazier, nothing like the 19th-century fantasy.

Your art: your palm trees from tm-mirage01 now replace mine in the desert. I cut them out of the drawing, removing the dune outlines and an arrowhead, and stored them as a compact 6.5 KB image. The clown stays as the observer.

Shimmer: light reaching the eye through the mirage now ripples slightly, as hot, turbulent air makes it do in all your photos. It's what makes the "water" read as water.

Checked:

- The lake and road scenes behave as intended, with and without the hot layer.
- In the layers panel, the stepped turning point converges to the smooth one as the layers increase (2.29 stack heights with 40 layers, 2.39 with 200, 2.42 smooth).
- All 44 controls work on a 360 px phone, and two copies on one MOM page stay independent.

One thing to watch: the MOM file is now 66.6 KB, mainly from the embedded images (your palms and the clown). Earlier you mentioned around 64 KB as a practical limit for another sim. If MOM objects, I can trim it, for example by storing the palms at a smaller size.

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